

SAE-Based Style Steering for Stable Diffusion 1.4

Methods, hyperparameters, and literature support

Every citation in this document was fact-checked against its primary source by independent adversarial verifiers (each claim faced three skeptics instructed to refute it). Claims that survived are cited below; the few that were contested are flagged inline. Checkmarks in the original review are omitted here for print clarity.

1. Methods and hyperparameters

Stage	Choice / value	Closest precedent
Base model	Stable Diffusion 1.4 (30 PNDM steps, CFG 7.5)	SAeUron (SD-1.5); ItD (SD-1.4)
Hookpoint	U-Net cross-attn up.1.2; d=1280, 16x16; cond CFG half only	SAeUron (identical)
SAE activation	TopK, k=32 (per-row eval / BatchTopK train)	OpenAI TopK-SAE; BatchTopK
Dictionary	expansion 16x -> 20480 latents (over-complete)	SAeUron (identical)
Dead-feature loss	AuxK, alpha=1/32, dead@10M tokens, k_aux~d/2	OpenAI TopK-SAE; Concept Steerers
Decoder	unit-norm rows, b_dec centering	Anthropic / OpenAI standard
Optimizer	Adam, lr 4e-4, batch 4096, 10 epochs	SAeUron (lr 4e-4); OpenAI (bs 4096)
Recon metric	FVU ~ 0.17 (normalized MSE = 1 - var explained)	OpenAI norm-MSE; SAeUron 0.198@up.1.2
Feature ID	contrast styled - neutral, keep >99.9th pct; patch: top-16/patch	SAeUron (concept-vs-complement, 99th pct)
Steering	ADD decoder dirs, cond half, every step; uniform vs patch	Concept Steerers (add); SAeUron (ablate)
Eval	CLIP ViT-B/32 + UnlearnCanvas ViT-L (51-way)	CLIPScore; UnlearnCanvas

2. Evaluation results (weights 1 - 2)

Means over 10 styles x 20 prompts; full cache in results/style_eval/scores.csv (2400 rows, uc_style filled on every row). Higher style/uc and lower content = stronger style at greater cost to the original content - the expected trade-off.

Condition	style_clip (up)	content_clip (up)	uc_style (up)
unstyled (baseline)	0.190	1.000	0.002
prompted (in X style)	0.249	0.772	0.012
injected_normal w1	0.200	0.938	0.008
injected_normal w2	0.218	0.876	0.020
injected_patches w1	0.203	0.932	0.006
injected_patches w2	0.224	0.858	0.011

Note: UC scores are low across all conditions because the UnlearnCanvas ViT-L classifier expects UnlearnCanvas's fine-tuned-SD appearance, not steered base SD-1.4 - worth a caveat in any writeup.

3. Why each choice is defensible

TopK / k=32, no L1 penalty

The k-sparse autoencoder "directly controls the number of active latents ... zeroing the rest," with loss "simply MSE" and no L1 penalty - removing the shrinkage bias where "the loss can be decreased by trading off some reconstruction accuracy for lower L1." TopK also yields a steeper MSE-vs-L0 frontier than ReLU. (One verifier contested the strong wording of the L1 claim, so treat "L1 always shrinks" as directional, not absolute.)

- Gao et al., *Scaling and Evaluating SAEs* (arXiv:2406.04093)
- Gated-SAE (arXiv:2404.16014)

BatchTopK (train) -> TopK (inference)

BatchTopK "relax[es] the top-k constraint to the batch level ... allowing a variable number of active latents per sample" and "largely achieves better NMSE ... for a fixed number of active latents of 32" than plain TopK. The canonical BatchTopK inference swaps to a JumpReLU global threshold; this pipeline uses per-row TopK instead - a deliberate simplification (the inference claim split verifiers 1-2, precisely because it is the recipe detail we diverge on).

- Bussmann et al., *BatchTopK SAEs* (arXiv:2412.06410)

AuxK, alpha=1/32, dead@10M

"An auxiliary loss (AuxK) similar to ghost grads ... models the reconstruction error using the top-k_aux dead latents ... scaled by a small coefficient alpha (usually 1/32)." Independently, Concept Steerers use the exact same "k_aux=256, loss weight alpha=1/32" for a diffusion SAE - strong support for the value. Without such mitigations "up to 90%" of latents go dead vs ~7% with them.

- Gao et al. (arXiv:2406.04093)
- Concept Steerers (arXiv:2501.19066)

16x expansion / over-complete dictionary

Over-completeness is foundational: Anthropic decomposed "a layer with 512 neurons into more than 4000 features," and SAE features are "more interpretable and monosemantic than directions identified by alternative approaches" such as PCA/ICA/neurons (verified 2-1).

- Bricken et al., *Towards Monosemanticity* (transformer-circuits.pub)
- Cunningham et al. (arXiv:2309.08600)

Hookpoint up.1.2 + conditional-only capture

A direct match to SAeUron: "For style filtering, we use second to last up-sampling block up.1.2 ... Importantly, we only gather feature maps related to text-conditioned generation part, discarding the unconditioned ones." SAeUron also confirms the exact SAE shape: "k=32 ... expansion factor of 16 ... n=20480 ... unit-normalized ... alpha=1/32." Intervening on the conditional half is grounded in CFG itself: the sampler "extrapolat[es] further in the direction of eps(x|y) and away from eps(x|empty)."

- Cywinski & Deja, *SAeUron* (arXiv:2501.18052)
- Ho & Salimans, *Classifier-Free Guidance* (arXiv:2207.12598)
- GLIDE (arXiv:2112.10741)

Feature ID by styled-vs-neutral contrast + percentile

SAeUron selects features by a normalized difference of "a concept's mean activation against the mean over the complement set" and filters by activation frequency at the "99th percentile." The styled-neutral contrast is the same machinery - but note the inversion: SAeUron contrasts one-style-vs-all-others for erasure (gamma<0 ablation), whereas this project contrasts styled-vs-neutral for addition. That is the genuine methodological deviation.

- Cywinski & Deja, *SAeUron* (arXiv:2501.18052)

Steering by addition of decoder directions

The additive form is the textbook activation-steering operator $R' = R \pm v$, with the direction from a contrast pair (ActAdd "Love"- "Hate"; CAA mean-difference). Most directly, Concept Steerers do the identical thing in a diffusion SAE: $x_{\text{steered}} = x + W_{\text{dec}}(\lambda * \text{ENC}(x_C))$, supporting both suppression and injection via the sign of λ . Caveat: SAE-TS shows it is "difficult to anticipate the effects of steering vectors ... [from] the direct use of SAE latents" - i.e. the strength sweep is the right way to handle that uncertainty.

- *RepE* ([arXiv:2310.01405](#))
- *ActAdd* ([arXiv:2308.10248](#))
- *CAA* ([arXiv:2312.06681](#))
- *Concept Steerers* ([arXiv:2501.19066](#))
- *SAE-TS* ([arXiv:2411.02193](#))

Evaluation: CLIP + UnlearnCanvas

CLIPScore is the canonical reference-free CLIP-similarity metric, "highest correlation with human judgments, outperforming CIDEr and SPICE" - though built for captioning, not style, so `style_clip` is an adaptation, not its designed use (flagged as a deviation). The UnlearnCanvas benchmark ships the "ViT-Large for style and object classification" used for `uc_style`.

- *Hessel et al., CLIPScore* ([arXiv:2104.08718](#))
- *Zhang et al., UnlearnCanvas* ([arXiv:2402.11846](#)); *OPTML-Group/UnlearnCanvas*

4. Bibliography (verified sources)

SAE foundations

- Gao et al. - Scaling and Evaluating Sparse Autoencoders (OpenAI TopK) - arXiv:2406.04093
- Bussmann, Leask, Nanda - BatchTopK Sparse Autoencoders - arXiv:2412.06410
- Cunningham et al. - SAEs Find Highly Interpretable Features in LMs - arXiv:2309.08600
- Bricken et al. - Towards Monosemanticity - transformer-circuits.pub/2023/monosemantic-features
- Rajamanoharan et al. - Improving Dictionary Learning with Gated SAEs - arXiv:2404.16014
- Rajamanoharan et al. - Jumping Ahead: JumpReLU Sparse Autoencoders - arXiv:2407.14435

Diffusion SAE / concept erasure

- Cywinski & Deja - SAeUron: Interpretable Concept Unlearning - arXiv:2501.18052
- Surkov et al. - One-Step is Enough: SAEs for T2I Diffusion - arXiv:2410.22366
- Concept Steerers: K-Sparse Autoencoders for Controllable Generation - arXiv:2501.19066
- ItD - SAE as Zero-Shot Classifier for Concept Erasing - arXiv:2503.09446
- Gandikota et al. - Erasing Concepts from Diffusion Models (ESD) - arXiv:2303.07345
- Gandikota et al. - Unified Concept Editing (UCE) - arXiv:2308.14761

Classifier-free guidance

- Ho & Salimans - Classifier-Free Diffusion Guidance - arXiv:2207.12598
- Dhariwal & Nichol - Diffusion Models Beat GANs (classifier guidance) - arXiv:2105.05233
- Nichol et al. - GLIDE - arXiv:2112.10741
- Bradley & Nakkiran - CFG is a Predictor-Corrector - arXiv:2408.09000
- CFG: From High-Dimensional Analysis to Generalized Guidance Forms - arXiv:2502.07849

Activation steering / representation engineering

- Zou et al. - Representation Engineering (RepE) - arXiv:2310.01405
- Turner et al. - Activation Addition (ActAdd) - arXiv:2308.10248
- Panickssery et al. - Steering Llama 2 via Contrastive Activation Addition (CAA) - arXiv:2312.06681
- SAE-TS - Improving Steering Vectors by Targeting SAE Features - arXiv:2411.02193

Evaluation

- Hessel et al. - CLIPScore - arXiv:2104.08718
- Zhang et al. - UnlearnCanvas - arXiv:2402.11846
- OPTML-Group/UnlearnCanvas - official code + ViT-Large classifiers - github.com/OPTML-Group/UnlearnCanvas